# Literature Review & Project Evaluation Framework

## Benchmark Study, Architectural Justification, Suggested Improvements, Lecturer Feedback & Grading Criteria

| Project Field | Project Specification Details |
|---|---|
| **Project Title** | EduHub — Full-Stack Online Learning Platform |
| **Initiative & Program** | Digital Egypt Pioneers Initiative (DEPI) — Software Engineering Track |
| **Document Ref / Code** | EDU-DOC-02 / DEPI-MS1 |
| **Document Version / State** | Version 1.0 (Official Milestone 1 Submission) |
| **Submission Due Date** | 16 October 2026 |
| **Architecture & Stack** | N-Tier Architecture \| ASP.NET Core MVC, EF Core, SQL Server, Azure, Stripe |
| **Project Team Members** | 1. Aya Abdelmoniem Mohamed Hamed<br>2. Ibrahim Mohamed Ibrahim Abdo<br>3. Reem Ashraf Mohammed Ahmed<br>4. Ahmed Ramadan Ali Mohamed<br>5. Warda Mahmoud Khalaf Ahmed |
| **Team Operating Model** | Cross-Functional Collaborative Team (Collective ownership across all components; all members actively participate in full-stack implementation, architecture, DB, and QA) |

## 1. Introduction & Evolution of Online Learning Platforms

Modern Electronic Learning (E-Learning) platforms and Massive Open Online Courses (MOOCs) have transformed education into an asynchronous, self-paced digital experience. This document presents a comprehensive Literature Review comparing leading industry platforms (Udemy, Coursera, Open edX, and Teachable) against the proposed EduHub architecture, followed by Suggested Project Improvements, the official Lecturer Feedback & Evaluation Rubric, and the Final Grading Criteria breakdown.

- **Generation 1 — Monolithic Institutional LMS (e.g., Moodle, Blackboard):** Designed for university administration and document sharing; suffers from dated UIs and complex scaling for cloud video streaming.
- **Generation 2 — Global MOOCs & Marketplaces (e.g., Udemy, Coursera, edX):** Introduced cloud video delivery, automated quizzes, and digital certificates, but impose high commissions or complex microservice overhead.
- **Generation 3 — Modular N-Tier Cloud Platforms (EduHub Approach):** Combines ASP.NET Core MVC server-side security and performance with cloud object storage (Azure Blob Storage) and trusted payment webhooks (Stripe).

## 2. Comparative Study of Existing Platforms & Market Gaps

| Evaluation Dimension | Udemy | Coursera | Open edX | Teachable | EduHub (Our Platform) |
|---|---|---|---|---|---|
| **Primary Model** | Open Commercial Marketplace | University & Enterprise Partner | Open-Source Institutional LMS | Creator SaaS Storefront | Moderated Marketplace & LMS |
| **Architecture** | Distributed Microservices | Proprietary Cloud Stack | Python/Django Microservices | Multi-Tenant Cloud SaaS | Modular N-Tier (ASP.NET Core) |
| **Course Structure** | Sections, Video Lessons, Quizzes | Weeks, Modules, Peer Review | Units, Subsections, XBlocks | Sections, Videos, Basic Quizzes | Sections, Ordered Lessons, Quizzes |
| **Pricing & Discounts** | Dynamic Promos & Coupons | Subscriptions & Financial Aid | Free Audit / Paid Verified Track | Creator Subscriptions & Coupons | Free/Paid + Strategy Pattern Coupons |
| **Media Security** | Encrypted HLS CDN Streaming | Protected Cloud Video CDN | S3 / External Video Hosts | Wistia / Cloudflare Stream | Private Azure Blob + Signed SAS URLs |
| **Payment Handling** | Multi-Gateway Marketplace | Stripe / Braintree Enterprise | Cybersource / E-Commerce Service | Stripe / PayPal SaaS Checkout | Stripe Checkout + Verified Webhooks |
| **Certification** | Standard Completion PDF | Identity-Verified Credential | Institutional Certificate | Custom Completion PDF | Dynamic Verifiable iTextSharp PDF |
| **Key Limitation Solved** | Retains up to 63% instructor revenue; lacks admin pre-review. | Closed to independent instructors; rigid academic schedules. | Very high DevOps & Kubernetes maintenance overhead. | High fixed monthly fees ($39–$299/mo) before any sales. | Clean N-Tier .NET stack, admin moderation, low overhead & secure media. |

## 3. Technical & Architectural Justification for EduHub

- **Three-Tier / N-Tier Separation (`EduHub.Web -> EduHub.Services -> EduHub.Data`):** Decouples HTTP controllers and Razor views from business rules and EF Core data access, allowing all 5 team members to work in parallel and write isolated unit tests.
- **Repository & Unit of Work Patterns:** Abstracts SQL Server queries behind interfaces (`ICourseRepository`, `IEnrollmentRepository`) and coordinates atomic multi-table database commits within a single transaction boundary.
- **Strategy Pattern for Pricing & Coupons:** Encapsulates regular pricing, percentage discounts, fixed-amount discounts, and promotional coupon validation behind a unified contract without brittle conditional blocks.

- **Observer Pattern for Domain Events:** Decouples key lifecycle events (`LessonCompleted`, `CourseCompleted`, `EnrollmentCreated`, `CoursePublished`) from notifications, progress recalculations, and idempotent certificate checks.

## 4. Suggested Improvements — Areas for Project Enhancement

Based on the literature review and architectural analysis, several strategic areas have been identified where EduHub can be enhanced both during iterative development and in post-MVP releases:

| Enhancement Area | Current Baseline (Initial Release) | Suggested Technical Improvement | Expected Impact & Benefit |
|---|---|---|---|
| **1. Video Processing & Adaptive Streaming** | Direct MP4 upload to private Azure Blob Storage with signed SAS URL playback in Video.js. | Integrate an automated FFmpeg / Azure Media processing pipeline for video compression, multi-bitrate HLS/DASH adaptive streaming, and automatic thumbnail extraction. | Reduces bandwidth consumption, eliminates buffering on slow mobile connections, and automates thumbnail generation. |
| **2. Automated Instructor Payouts & Split Billing** | Payments collected via Stripe Checkout with database tracking of course revenue and instructor earnings. | Upgrade to Stripe Connect (Express/Custom accounts) to automate revenue splitting, platform fee deduction, and scheduled bank payouts to instructors. | Provides full end-to-end marketplace financial automation and compliance. |
| **3. Real-Time Student Q&A & Notifications** | Standard server-rendered MVC announcements, course questions, and database notifications. | Integrate ASP.NET Core SignalR WebSockets for real-time instant notifications, live Q&A discussion threads, and interactive study rooms. | Increases student engagement and reduces instructor response latency. |
| **4. Caching & High-Concurrency Search** | SQL Server indexed queries through EF Core for course browsing, filtering, and category navigation. | Introduce Redis Distributed Caching (`IDistributedCache`) for hot catalog pages and Azure AI Search / full-text indexing for instant typo-tolerant search. | Cuts catalog database load by >70% and delivers sub-50ms search responses at scale. |
| **5. Enhanced Certificate Verification (QR Code)** | PDF certificate generated via iTextSharp with unique `CertificateNumber` stored in SQL Server. | Embed a dynamic QR Code on every generated PDF certificate linking directly to a public verification endpoint (`/Certificates/Verify/{code}`). | Enables employers and universities to verify certificate authenticity instantly via smartphone scan. |
| **6. Multi-Course Shopping Cart & Bundles** | Direct single-course enrollment and Stripe purchase flow with single coupon application. | Implement a persistent Shopping Cart (`CoursePurchase` linking entity) supporting multi-course checkout and course bundle discounts. | Increases average order value (AOV) and improves checkout convenience for students. |
| **7. AI-Assisted Learning & Recommendations** | Manual search, category filtering, and static related-course suggestions. | Integrate semantic course recommendations based on `StudentProfile.LearningPreferences` and automated quiz explanation hints. | Personalizes the learning path and improves course completion rates. |

## 5. Final Grading Criteria — Breakdown of Marks

The final project grade (100 Marks total) is evaluated across four primary pillars: Documentation (20%), Implementation & Execution (40%), Testing & Security QA (20%), and Final Presentation & Defense (20%).

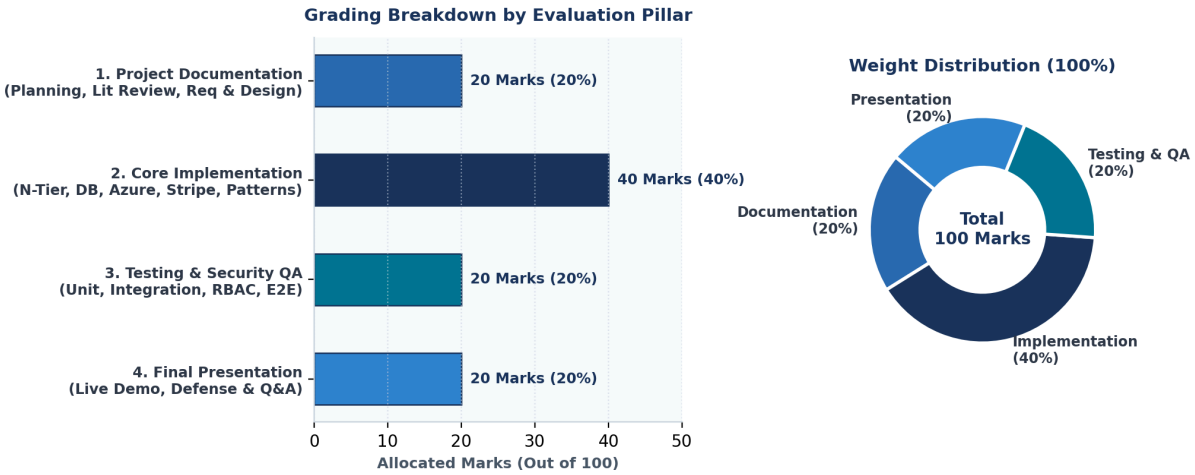



*Figure 1: EduHub Final Grading Criteria & Mark Allocation Breakdown (100 Marks Total)*